

A PROOF OF THE APPLGATE–LAGARIAS CONJECTURE A AND THE $3x + 1$ GROWTH EXPONENT CONJECTURE

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ABSTRACT. For an integer a let $\pi_a(x)$ be the number of integers n with $|n| \leq x$ whose trajectory under the $3x + 1$ map T , given by $T(n) = n/2$ for even n and $T(n) = (3n + 1)/2$ for odd n , contains a . In 1995 Applegate and Lagarias conjectured that for every a not divisible by 3 there is $c_a > 0$ with $\pi_a(x) \geq c_a x$ for all $x \geq |a|$ (Conjecture A). We prove this conjecture. For positive targets the input is a recently released theorem, formally verified in Lean: the predecessors of each positive target prime to 3 under the ordinary map $n \mapsto n/2$, $3n + 1$ have positive lower density. Negative targets require new mathematics. Negation conjugates the $3x + 1$ dynamics on the negative integers to the $3x - 1$ dynamics on the positive integers, and we prove the corresponding density theorem for $3x - 1$. Its proof follows the architecture of the proof of that theorem. The sign change reverses one basic inequality, which costs a multiplicative factor at every generation of the inverse construction. We show that the product of these factors stays uniformly bounded. Ordinary and accelerated predecessor sets can differ only at targets $a \equiv 4 \pmod{6}$, where $T(2a) = a$ transfers the bound, and the target’s membership in its own predecessor set gives the bound for every $x \geq |a|$. As a corollary, $\log \pi_a(x) / \log x \rightarrow 1$, which proves the $3x + 1$ growth exponent conjecture. The complete argument, including the imported theorem, is formalized in Lean 4 with Mathlib and has been replayed independently by the Lean kernel.

1. INTRODUCTION

1.1. Inverse iterates of the $3x + 1$ map. Let $T : \mathbb{Z} \rightarrow \mathbb{Z}$ be the $3x + 1$ map in the form used by Applegate and Lagarias [1, (1.1)],

$$T(n) = \begin{cases} (3n + 1)/2, & n \text{ odd,} \\ n/2, & n \text{ even.} \end{cases}$$

The $3x + 1$ conjecture asserts that every positive integer eventually reaches 1. A way to measure what is known is to fix a target a and count the integers whose trajectories pass through it. Following [1, (1.2)], set

$$(1) \quad \pi_a(x) = \#\{n \in \mathbb{Z} : |n| \leq x \text{ and } T^{(k)}(n) = a \text{ for some } k \geq 0\}.$$

If $3 \mid a$, the only predecessors of a are the numbers $2^j a$, so $\pi_a(x)$ grows logarithmically. For $a \not\equiv 0 \pmod{3}$ the count grows at least like a power of x : the exponent 0.65 was obtained in [1, Theorem 1.2], where the earlier bounds are surveyed [1, §1], and the exponent 0.84 is recorded in [4, Theorem 2.5]. Applegate and Lagarias conjectured that the growth is in fact linear.

Conjecture 1.1 (Applegate–Lagarias [1, Conjecture A]). *For each $a \not\equiv 0 \pmod{3}$ there is a positive constant c_a such that $\pi_a(x) \geq c_a x$ for all $x \geq |a|$.*

Date: September 2026.

2020 Mathematics Subject Classification. Primary 11B83; Secondary 37P99, 11B37, 68V20.

Key words and phrases. $3x + 1$ problem, Collatz problem, $3x - 1$ problem, inverse iterates, predecessor density, Applegate–Lagarias conjectures, growth exponent, formal verification.

Kontorovich and Lagarias [4, Definition 2.11] define the upper and lower growth exponents

$$\eta_3^+(a) = \limsup_{x \rightarrow \infty} \frac{\log \pi_a(x)}{\log x}, \quad \eta_3^-(a) = \liminf_{x \rightarrow \infty} \frac{\log \pi_a(x)}{\log x},$$

and record the following conjecture, which they attribute to Applegate and Lagarias and note is weaker than Conjecture 1.1.

Conjecture 1.2 ($3x + 1$ growth exponent conjecture [4, Conjecture 2.1]). *For every integer $a \not\equiv 0 \pmod{3}$ the growth exponent exists and equals one: $\eta_3^+(a) = \eta_3^-(a) = 1$.*

Both conjectures are predicted by stochastic models of the inverse iteration [4].

1.2. Main results.

Theorem A (Conjecture A). *For every $a \in \mathbb{Z}$ with $3 \nmid a$ there is $c_a > 0$ such that $\pi_a(x) \geq c_a x$ for every real $x \geq |a|$.*

Theorem B (Growth exponents). *For every $a \in \mathbb{Z}$ with $3 \nmid a$,*

$$\lim_{x \rightarrow \infty} \frac{\log \pi_a(x)}{\log x} = 1.$$

In particular $\eta_3^+(a) = \eta_3^-(a) = 1$, which proves Conjecture 1.2.

The negative targets of Theorem A rest on a density theorem for the $3x - 1$ map $C_-(m) = m/2$ for even m , $C_-(m) = 3m - 1$ for odd m .

Theorem C (Predecessor density for $3x - 1$). *For every integer $a \geq 1$ with $3 \nmid a$, the set $\{m \geq 1 : C_-^k(m) = a \text{ for some } k \geq 0\}$ has positive lower natural density.*

1.3. From the positive-target theorem to Conjecture A. The positive-target input is the predecessor-density theorem of [6], stated as Theorem 2.2 in Section 2. It differs from Conjecture A in four respects.

	Theorem 2.2	Conjecture A	What is needed
map	ordinary C_+	accelerated T	elementary except at $a \equiv 4 \pmod{6}$; Section 4
target	$a \geq 1$	$a \in \mathbb{Z}$	new theorem for $3x - 1$; Sections 3, 5
starting points	$n \geq 1$	$n \in \mathbb{Z}$	embedding, and negation for $a < 0$
cutoff	integer $X \geq X_a$	every real $x \geq a $	elementary; Section 6

Three of the four gaps are closed by elementary arguments. The second is not. The dynamics of T on the negative integers is the $3x - 1$ dynamics in disguise (Lemma 5.1), and no density theorem for $3x - 1$ was available.

1.4. Contributions and relation to prior work.

- (i) *Theorem C* (Section 3). It is proved by running the argument of [5, 6] for the $3x - 1$ map. We identify the four places where the sign of the ± 1 enters. Residue conditions transfer exactly under the negation $z \mapsto -z$ of 3-adic residues (Lemma 3.2). The one inequality that reverses is the charge inequality $x_w \omega(w) \leq R$ between a source, its weight and its root. It is repaired by a sharp charge bound (Lemma 3.4) whose excess is summable along the scale recursion, so the accumulated loss is bounded by an absolute constant $\mathcal{E} < 1 + 5 \cdot 10^{-50}$ (Lemma 3.5).
- (ii) *The signed assembly* (Sections 4–6). The conjugacy $C_+(-m) = -C_-(m)$ carries Theorem C to negative targets. An ordinary-to-accelerated bridge handles the map, and a threshold argument gives the exact form of Conjecture A. For positive targets the bridge is already noted in [6, §1.2]: Theorem 2.2, applied to $2a$, yields positive density for T .

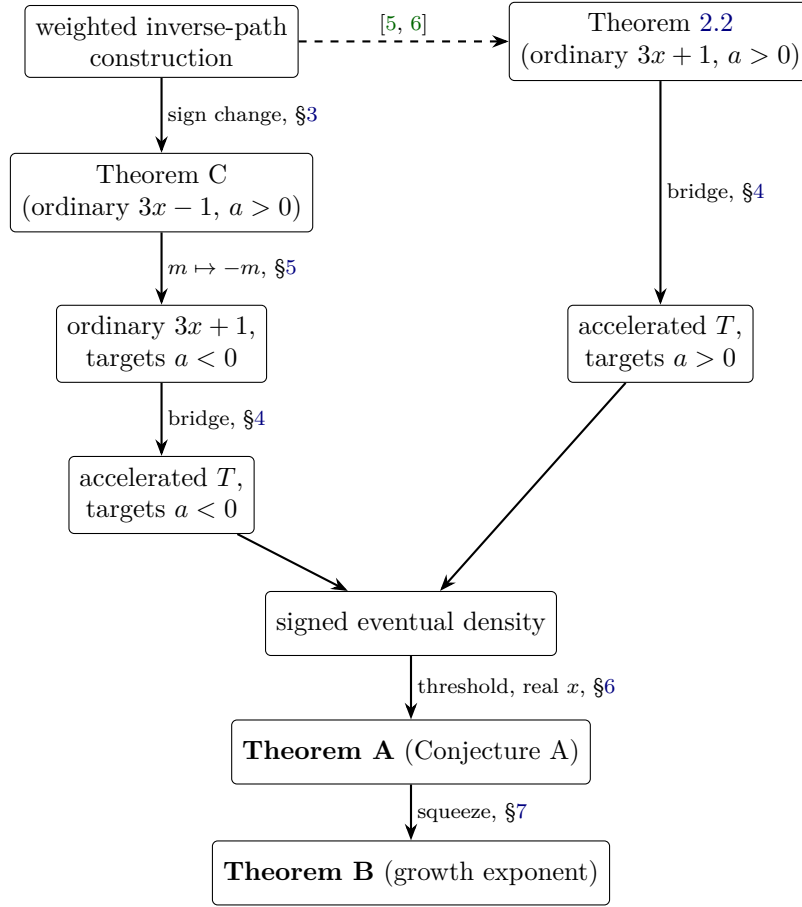


FIGURE 1. Structure of the proof. The dashed arrow is the imported theorem, proved in [5, 6]. Theorem C reuses the construction behind Theorem 2.2, not Theorem 2.2 itself.

(iii) *Theorem B* (Section 7), a short consequence of Theorem A.

Theorem 2.2 is the only imported result. Its human-readable proof is in the manuscripts [5, 6], which have not been refereed. Section 2.3 records their provenance and Section 9 discusses what it implies.

1.5. The formal proof. Theorems A, B and C, together with Theorem 2.2 and every lemma used to derive them, are formally proved in Lean 4 [2] with Mathlib [7]. The development depends only on Lean’s three standard axioms, and the Lean kernel has independently replayed it. Both conjectures are restated in Lean from first principles and discharged by the main theorems, which checks that the formal statements are the published ones. Section 8 gives the details.

1.6. Organization. Figure 1 shows the structure of the proof. Section 2 sets up notation, states Theorem 2.2 and describes the steps of its proof. Section 3 proves Theorem C. Sections 4 and 5 pass to the accelerated map and to negative targets. Section 6 proves Theorem A and Section 7 proves Theorem B. Section 8 describes the formal verification, Section 9 discusses the constants and the scope of the result, and Section 10 lists further questions.

2. PRELIMINARIES AND THE INPUT THEOREM

2.1. Maps, reachability, density. Besides T we use the ordinary maps

$$C_+(n) = \begin{cases} 3n+1, & n \text{ odd}, \\ n/2, & n \text{ even}, \end{cases} \quad C_-(n) = \begin{cases} 3n-1, & n \text{ odd}, \\ n/2, & n \text{ even}, \end{cases}$$

with C_+ defined on $\mathbb{Z}$ and C_- on the positive integers, which it preserves. For a map f write $n \rightsquigarrow_f a$ if $f^k(n) = a$ for some $k \geq 0$; the case $k = 0$ is allowed, so $a \rightsquigarrow_f a$. For a set A of positive integers and $X \in \mathbb{N}$ let $A(X) = \#\{n \in A : n < X\}$. The set A has *positive lower density* if there are $c > 0$ and X_0 with $A(X) \geq cX$ for every integer $X \geq X_0$. We write $G_k = \mathbb{Z}/3^k\mathbb{Z}$ and $\langle f \rangle_k = 3^{-k} \sum_{z \in G_k} f(z)$ for $f: G_k \rightarrow \mathbb{R}$.

2.2. Syracuse maps and exact words. For a positive odd integer x put

$$S_+(x) = \frac{3x+1}{2v_2(3x+1)}, \quad S_-(x) = \frac{3x-1}{2v_2(3x-1)}.$$

Both are maps of the positive odd integers to themselves. The ordinary orbit of an odd x under $C_\pm$ passes through $S_\pm(x)$. Hence $x \rightsquigarrow_{S_\pm} R$ implies $x \rightsquigarrow_{C_\pm} R$.

Definition 2.1. Let R be a positive odd integer. A finite sequence $w = (a_1, \dots, a_d)$ of positive integers is an *exact word for $S_\pm$ ending at R* if there is a positive odd integer x_w , the *source*, with $S_\pm^d(x_w) = R$ and $a_i = v_2(3S_\pm^{i-1}(x_w) \pm 1)$ for $1 \leq i \leq d$. We write $d = d(w)$, $A(w) = a_1 + \dots + a_d$, and define the *weight* and the *offset*

$$\omega(w) = 3^d 2^{-A(w)}, \quad B_w = \sum_{i=1}^d 3^{d-i} 2^{a_1 + \dots + a_{i-1}}.$$

The weight is the one used in [5, (2.1)] and [6, (3.1)]. The weight and the offset depend only on the word, not on the sign. Iterating $3x \pm 1 = 2^a S_\pm(x)$ along the orbit gives, by induction on d ,

$$(2) \quad 3^d x_w \pm B_w = 2^{A(w)} R, \quad \text{equivalently} \quad x_w \omega(w) = R \mp \frac{B_w}{2^{A(w)}}.$$

For $3x+1$ this gives the *charge inequality* of [6, (3.2)],

$$(3) \quad x_w \omega(w) \leq R \quad (\text{sign } +).$$

For $3x-1$ the same identity gives $x_w \omega(w) \geq R$ instead.

2.3. The input theorem.

Theorem 2.2 (Positive density of $3x+1$ predecessors [6, Theorem 1.1]). *For every integer $a \geq 1$ with $3 \nmid a$, the set $P(a) = \{n \geq 1 : n \rightsquigarrow_{C_+} a\}$ has positive lower density.*

The statement allows even targets and assumes neither that a reaches 1 nor that a is non-periodic. Its constants depend on a .

Provenance. Theorem 2.2 was released in September 2026 on the ProofAtlas platform, as the manuscript [6] with a Lean development built on the library of [5]. The manuscripts state that the research was directed by their author and that generative AI (OpenAI models used through the ProofAtlas harness) contributed substantially to the mathematical exploration, the proof development, the formalization and the exposition. Neither manuscript reports human peer review. We therefore do not take the correctness of Theorem 2.2 on the authority of its exposition. It rests on the formal proof, which we replayed independently in a fresh kernel environment (Section 8.5). The manuscripts serve here as the human-readable description of the proof's structure.

2.4. The structure of the proof of Theorem 2.2. The proof in [6] combines five steps. Steps (P1), (P3), (P4) and (P5) are proved in [6]. Step (P2) is assembled there from the estimates of [5], in its Appendix A.

- (P1) *Seeds* [6, §2]. For every modulus 3^q , every residue y and every L there is a positive odd $R > L$ with $R \equiv y \pmod{3^q}$, $R \rightsquigarrow_{C_+} a$, and no positive return, that is, $S_+^t(R) \neq R$ for all $t \geq 1$. The seeds come from the explicit family $R_k = ((3r_0 + 1)4^k - 1)/3$, which runs through all residues modulo 3^q . No return for large R comes from a lemma valid for every map $f: \mathbb{N} \rightarrow \mathbb{N}$ [6, Lemma 2.1].
- (P2) *A residue-weighted mass* [6, Proposition 3.1 and Appendix A], [5, §§2–6, 8]. Starting from a seed, inverse histories are built generation by generation. At generation j one appends an exact word that first crosses a prescribed linear barrier at scale b_j , with bounded overshoot, where $b_0 = 2^{80}$ and $b_{j+1} = b_j + \lfloor b_j/100 \rfloor$. The endpoint of each history becomes a root for the next generation. Each history is paired with a reference density $\rho_k: G_k \rightarrow [0, \infty)$ evaluated at its current endpoint. This density comes from the 3-adic random walk $Y_{k+1} = 2^{-E_{k+1}}(3Y_k + 1)$ with independent geometric E_j , so it has mean $2/3$ and vanishes on nonunits. A finite-distribution mixing estimate gives $\langle |\rho_k - \rho_m \circ \pi_{k,m}| \rangle_k \leq C/m^2$ with an explicit constant C [5, Lemma 8.1]. Averaging over residues selects a unit residue y at which the initial mark is at least $255/256$. Summable variation estimates show that the mark stays at least $175/256 > 2/3$ at every late generation, uniformly in the external scale X . Terminal sources lie in $[X, 32X)$.
- (P3) *Distinct sources* [6, Lemma 4.1]. If the seed has no positive return, two words ending at it with the same source must have the same length, and hence coincide. So $w \mapsto x_w$ is injective, and by (3) the words contribute weight at most R/X to each distinct source in $[X, \infty)$.
- (P4) *Removing the residue weight* [6, §4.2]. A residue-class count and the mixing estimate at a fixed coarse modulus 3^m , with $m = 132RC + 1$, turn the weighted mass into a lower bound $U \geq \eta = 3/(8 \cdot 3^m)$ for the total weight of the selected words.
- (P5) *Counting* [6, §4.3]. Steps (P3) and (P4) give at least XU/R distinct predecessors in $[X, 32X)$. This yields Theorem 2.2 with $c_a = 3/(256 R 3^m)$ and $X_a = 32(H + 1)$ for a height H depending on R .

The sign of the ± 1 enters this proof at four points. We use this list in Section 3.

- (S1) the Syracuse map itself, hence which words are exact and what the seeds are (P1);
- (S2) the affine identity (2), through the congruence that makes a word exact and through the residue maps and transfer operators of [5, §2] (P2);
- (S3) the charge inequality (3) and the block-size bounds derived from it, [5, (3.5)], which enter the histogram and variation estimates (P2), the source window (P2) and the count (P3), (P5);
- (S4) the reduction of a terminal valuation by even amounts, which acts on sources by the affine map $\Psi_j(x) = 4^j x + (4^j - 1)/3$ (P2), (P4).

The rest is independent of the sign. This includes the combinatorics of words (first crossings, overshoot and length restrictions), their geometric probabilities, the mixing estimate, the summable variation majorants, the scale-coverage argument and the non-return lemma.

3. POSITIVE PREDECESSOR DENSITY FOR THE $3x - 1$ MAP

We prove Theorem C by running the proof of Theorem 2.2 for S_- and C_- . We do not reprove the sign-independent estimates of [5, 6]; we prove the statements that replace the four sign-sensitive points (S1)–(S4) and track the constants they change.

3.1. Exact words for $3x - 1$. For S_- the identity (2) reads $3^d x_w = 2^{A(w)} R + B_w$. Unlike the $3x + 1$ case, no condition is needed to keep the source positive.

Lemma 3.1 (Decoding). *Let R be a positive odd integer and $w = (a_1, \dots, a_d)$ a word of positive integers. If $3^d \mid 2^{A(w)} R + B_w$, then w is an exact word for S_- ending at R , with source $x_w = (2^{A(w)} R + B_w)/3^d$.*

Proof. Define $y_d = R$ and $y_{i-1} = (2^{a_i} y_i + 1)/3$. These are rationals whose denominators are powers of 3, and $y_0 = (2^{A(w)} R + B_w)/3^d$ by the computation behind (2), so y_0 is an integer by hypothesis. If y_i were not an integer, then $v_3(y_i) < 0$ and $v_3(y_{i-1}) = v_3(y_i) - 1 < 0$, since $v_3(2^{a_i} y_i + 1) = v_3(y_i)$. Inductively y_0 would not be an integer. Hence every y_i is an integer, and by downward induction every y_i is positive. Since $2^{a_i} y_i + 1$ is odd, so is y_{i-1} . Finally $3y_{i-1} - 1 = 2^{a_i} y_i$ with y_i odd, so $v_2(3y_{i-1} - 1) = a_i$ and $S_-(y_{i-1}) = y_i$. $\square$

3.2. The negation correspondence. A word is exact for S_- at R if and only if $2^{A(w)} R \equiv -B_w \pmod{3^d}$. A word is exact for S_+ at R' only if $2^{A(w)} R' \equiv B_w \pmod{3^d}$. So the $3x - 1$ condition at R is the $3x + 1$ condition at the residue $-R$. The next lemma makes this precise for every residue statement, not only for exactness.

Lemma 3.2 (Negation correspondence). *Let w be a word, and let x, R be integers with $3^d x = 2^{A(w)} R + B_w$. For every integer t the integers*

$$x' = 2^{A(w)} t - x, \quad R' = 3^d t - R$$

satisfy $3^d x' + B_w = 2^{A(w)} R'$. If $3^Q \mid t$ and t is large, then $x' \equiv -x$ and $R' \equiv -R \pmod{3^Q}$, and $x', R' > 0$.

Proof. $3^d x' + B_w = 3^d 2^{A(w)} t - (2^{A(w)} R + B_w) + B_w = 2^{A(w)} (3^d t - R)$. The other claims are immediate. $\square$

Every statement of [5, 6] that depends on a pair (source, root) only through their residues modulo powers of 3 and through the $3x + 1$ affine relation therefore holds for $3x - 1$ pairs after both residues are negated. We use three instances.

Corollary 3.3. (a) *Residue maps. Let F_w be the injective residue map of [5, §2] attached to w (in [5] the letters are listed from the root outwards). The corresponding map for S_- is $z \mapsto -F_w(-z)$, and the transfer identity $\langle |T_w g| \rangle = 2^{-A(w)} \langle |g| \rangle$ of [5, Lemma 2.1] holds unchanged.*

(b) *The mass at the negated residue. Fix a scale and a word filter ψ , and let $g: G_k \rightarrow \mathbb{R}$ vanish on nonunits. The weighted sum $\sum_w \omega(w) \psi(w) g(-x_w)$ over the selected exact S_- -words ending at R equals the value at $-R$ of the residue kernel through which [5] computes the corresponding $3x + 1$ sum.*

(c) *The recovered digit. In the regeneration step, the ternary digit attached to a child with source x is $-x \bmod 3$ rather than $x \bmod 3$.*

Proof. Each item follows by applying Lemma 3.2 to the pairs (source, root) that occur in the corresponding definition of [5]. The Lean statements are listed in Appendix A. $\square$

In (b) the reference density is evaluated at $-x_w$. Negation is a bijection of G_k that preserves units and uniform means, so the mean, the vanishing on nonunits and the mixing estimate of the reference densities hold for the negated densities as stated. We do not introduce a new reference law. The $3x - 1$ random walk $Y_{k+1}^- = 2^{-E_{k+1}} (3Y_k^- - 1)$ is the negative of the walk of [5, §2], and the correspondence says that using that law at the negated residue is the same thing.

3.3. The inequalities that reverse. Let $x = N_0 \rightarrow N_1 \rightarrow \dots \rightarrow N_d = R$ be the S_- -orbit of a source, with valuations a_i and partial sums $A_i = a_1 + \dots + a_i$. Put $\Pi_i = 3^i N_0 / (2^{A_i} N_i)$. Then $\Pi_0 = 1$ and $\Pi_d = x_w \omega(w) / R$. The charge inequality reverses, and it does so in three forms.

(R1) *Step by step.* From $3N_{i-1} \pm 1 = 2^{a_i} N_i$ we get

$$\Pi_i = \Pi_{i-1} \cdot \frac{3N_{i-1}}{3N_{i-1} \pm 1}.$$

For $3x + 1$ the factor is below 1, so Π is non-increasing and $\Pi_d \leq 1$, which is (3). For $3x - 1$ the factor is $1 + 1/(3N_{i-1} - 1) > 1$.

(R2) *The charge.* Consequently $x_w \omega(w) = R + B_w 2^{-A(w)} \geq R$ for $3x - 1$.

(R3) *Block sizes.* For a selected word of length s and valuation sum W with root M , the room condition $2 \cdot 3^s < M$ used in [5, (3.5)] gives for $3x + 1$

$$2^W M \leq 2 \cdot 3^s N, \quad 3^s N \leq 2^W M,$$

where N is the source. For $3x - 1$ the offset bound $B_w \leq 2^W 3^s$ and the same room condition give instead

$$2^W M \leq 3^s N \leq 2 \cdot 2^W M.$$

The lower bound improves and the upper bound worsens by a factor 2.

Why the reversal is dangerous. The proof of Theorem 2.2 uses (3) with constant exactly 1 once per generation. It enters the telescoped charge of a history, the weight bound $\omega(w) 16^{\lfloor b/100 \rfloor} \leq 1$ of a block, and through these the residue histogram bound [5, (3.9)] that controls every variation estimate. The number of generations needed to reach scale X tends to infinity with X . A fixed constant $\kappa > 1$ per generation would produce κ^n and destroy the uniformity in X on which the proof rests. Telescoping (R1) with a lower bound F for the orbit values gives $(1 + 1/(3F - 1))^d$ with d of order $\log X$. That bound also fails: the seed, and hence F , is fixed before X is chosen, so the density constant would decay like $1/\log X$. What is needed is a per-generation excess that is summable.

3.4. The sharp charge bound.

Lemma 3.4 (Sharp charge). *Let w be an exact word for S_- ending at R that is selected at scale $b \geq 200$, so that $d(w) \leq b + \lfloor 3b/5 \rfloor \leq 8b/5$ [5, §3], and suppose $R \geq 16^b$. Then*

$$x_w \omega(w) \leq (1 + 3(9/16)^b) R.$$

If moreover the source is at least $16^{\lfloor b/100 \rfloor} R$, as it is for the blocks of the construction [5, (3.5)], then $\omega(w) 16^{\lfloor b/100 \rfloor} \leq 1 + 8^{-b}$.

Proof. Write $A = A(w)$, $d = d(w)$ and $A_i = a_1 + \dots + a_i$. Since every $a_j \geq 1$, we have $A_{i-1} \leq A - (d - i + 1)$, so

$$B_w = \sum_{i=1}^d 3^{d-i} 2^{A_{i-1}} \leq 2^A \sum_{i=1}^d \frac{3^{d-i}}{2^{d-i+1}} \leq 2^A (3/2)^d \leq 2^A 3^d.$$

Hence $x_w \omega(w) - R = B_w 2^{-A} \leq 3^d \leq 3 \cdot 9^b = 3(9/16)^b 16^b \leq 3(9/16)^b R$. For the second claim, $\omega(w) 16^{\lfloor b/100 \rfloor} \leq \omega(w) x_w / R = 1 + B_w / (2^A R)$. The middle bound above gives $B_w 2^{-A} \leq (3/2)^d \leq (3/2)^{8b/5} < 2^b$, since $(3/2)^8 < 2^5$; this is the single-block offset bound of [5, proof of Proposition 3.1]. With $R \geq 16^b$ we get $1 + B_w / (2^A R) \leq 1 + 8^{-b}$. $\square$

For comparison, the constant is exactly 1 for $3x + 1$. The crude form of the same argument, using only $2 \cdot 3^d < R$, gives $x_w \omega(w) \leq \frac{3}{2} R$. That constant is harmless if paid once, but it compounds over generations.

3.5. Iterating generations.

Lemma 3.5 (Uniform product). *Let $b_0 \geq 200$ and $b_{k+1} \geq b_k + 2$ for all k , and let $0 \leq \varepsilon_k \leq 3(9/16)^{b_k}$. Then for every n*

$$\prod_{k < n} (1 + \varepsilon_k) \leq \mathcal{E} := \exp\left(3 \cdot \frac{256}{175} \cdot (9/16)^{200}\right) < 1 + 5 \cdot 10^{-50}.$$

Proof. $b_k \geq 200 + 2k$, so $\varepsilon_k \leq 3(9/16)^{200}(81/256)^k$ and $\sum_k \varepsilon_k \leq 3(9/16)^{200}/(1 - 81/256) = 3 \cdot \frac{256}{175}(9/16)^{200}$. Use $1 + t \leq e^t$. Numerically $(9/16)^{200} < 1.1 \cdot 10^{-50}$. $\square$

The scales of [5, §3] satisfy the hypothesis: $b_0 = 2^{80}$ and $b_{k+1} = b_k + \lfloor b_k/100 \rfloor \geq b_k + 2$. Applying Lemma 3.4 once per generation and Lemma 3.5 to the product gives the $3x - 1$ versions of the facts that the proof of Theorem 2.2 uses with constant 1.

Corollary 3.6. *In the $3x - 1$ construction, for every generation n :*

- (i) *a terminal source x of total weight W descending from the seed R satisfies $xW \leq \mathcal{E}R$;*
- (ii) *the product of block weights along a history satisfies $W 16^{b_n} \leq \mathcal{E} 16^{b_0}$;*
- (iii) *the history affine identity gives $M \leq WR_n \leq M + 2^{b_0+1}$ for the endpoint R_n of a history from M . For $3x + 1$ it gives $M - 2^{b_0+1} \leq WR_n \leq M$; the band has the same width.*

Part (iii) is why the residue histogram bound [5, (3.9)] survives the change of sign without loss. It only uses the width of the band. Parts (i) and (ii) multiply every capacity, variation, terminal and loss majorant of [5, §§3, 5, 8] by at most $\mathcal{E}$.

3.6. Seeds.

Lemma 3.7 (Seeds for $3x - 1$). *Let $a \geq 1$ with $3 \nmid a$. For every $q \geq 0$, every residue $y \pmod{3^q}$ and every L , there is a positive odd integer $R > L$ with $R \equiv y \pmod{3^q}$, $R \rightsquigarrow_{C_-} a$, and $S_-^t(R) \neq R$ for all $t \geq 1$.*

Proof. Choose $e \in \{1, 2\}$ with $2^e a \equiv -1 \pmod{3}$ and put $r_0 = (2^e a + 1)/3$. For $k \geq 0$ let

$$R_k = \frac{(3r_0 - 1)4^k + 1}{3}, \quad \text{so that} \quad 3R_k - 1 = 2^{2k+e}a.$$

Each R_k is a positive odd integer, and $C_-^{1+2k+e}(R_k) = a$. Since $R_{k+h} - R_k = 2^e a 4^k (4^h - 1)/3$ with $3 \nmid 2^e a$, and $v_3(4^h - 1) = 1 + v_3(h)$, we get $v_3(R_{k+h} - R_k) = v_3(h)$. Hence $R_0, \dots, R_{3^q-1}$ are distinct modulo 3^q and cover every residue, and adding multiples of 3^q to the index preserves the residue. All R_k with $k \geq 1$ have the same Syracuse image, the odd part u of a . By [6, Lemma 2.1] applied to $f = S_-$ and the endpoint u , all sufficiently large R_k have no positive return. $\square$

This is the $3x - 1$ analogue of [6, Lemma 2.3]. The cycles of S_- on the positive integers, namely $\{1\}$, $\{5, 7\}$ and $\{17, 25, 37, 55, 41, 61, 91\}$, play no role. As in [6, Lemma 4.1], the no-return property is exactly what the distinct-source argument of (P3) needs.

3.7. Proof of Theorem C. Fix $a \geq 1$ with $3 \nmid a$. We run (P1)–(P5) for S_- with the following changes.

Seed class (P1), (P2). The averaging argument of [5, §8.3] and [6, Appendix A.2] depends only on residues. At the fixed stage and modulus it yields a unit residue y at which the pulled-back reference mark is at least $\prod_j p_j \geq 255/256$. By Corollary 3.3(b), the mark of $3x - 1$ histories rooted at R is the mark of [5] at $-R$. By Lemma 3.7 we choose a seed $R \geq 16^{b_0}$ with $R \equiv -y \pmod{3^q}$, $R \rightsquigarrow_{C_-} a$ and no positive return. The class $-y$ is a unit class because y is.

Surviving mass (P2). In [5, §8.3] three errors are bounded by $1/16$, $1/8$ and $1/8$: the central variation, the terminal comparison and the excluded terminal mass. Each bound is a sum of majorants built from the residue histogram and a kernel mean. By Corollary 3.6 each majorant is multiplied by at most $\mathcal{E}$. By Corollary 3.3(b) the kernel means are unchanged. Using only $\mathcal{E} \leq 2$, the mark stays at least $255/256 - 2 \cdot 5/16 = 95/256 \geq 1/4$ at every late generation, uniformly in the admissible scale X . For $3x + 1$ the corresponding bound is $175/256 \geq 2/3$ [5, (8.8)].

Window (P2). By (R3) with overshoot one, terminal sources lie in $[X, 64X)$ instead of $[X, 32X)$. The scale-coverage argument of [5, Lemma 6.2] still applies. The extra factor 2 raises the per-generation spread by one unit, which stays quadratic in the generation index, while the scales grow geometrically.

Removing the residue weight (P4). The residue-class count of [6, §4.2] is applied on $[X, 64X)$, where each class modulo $3^k \leq X$ contains at most $66X/3^k$ integers, and with the charge constant $\mathcal{E}$. The mixing error term becomes at most $88\mathcal{E}RC/m^2$, compared with $44RC/m^2$ for $3x + 1$. Choosing $m = 1408[R]C + 1$ makes it at most $1/8$. The coarse term is at most $\frac{8}{9}3^mU$ as before. Hence

$$U \geq \eta^- := \frac{9}{64 \cdot 3^m}.$$

Counting (P3), (P5). The seed has no positive return, so $w \mapsto x_w$ is injective, and by Corollary 3.6(i) each distinct source $x \geq X$ receives weight at most $\mathcal{E}R/X$. Every source reaches R under S_- , hence reaches a under C_- . For every integer $Y \geq 64(H + 1)$, taking $X = Y/64$ gives

$$\#\{1 \leq n < Y : n \rightsquigarrow_{C_-} a\} \geq \frac{XU}{\mathcal{E}R} \geq \frac{\eta^-}{64\mathcal{E}R} Y.$$

Here H bounds the admissible scales of a fixed generation. It exists because that generation has finitely many histories. This proves Theorem C with $c_a^- = 9/(4096\mathcal{E}R3^m)$ and $X_a^- = 64(H + 1)$. $\square$

Remark 3.8 (The formal proof of Theorem C). Theorem C is formally proved as `ThreeXMinusOne.predecessors_positive_lower_density`. The formal proof re-proves the sign-sensitive statements of this section and uses the sign-independent lemmas of [5, 6] unchanged. It does not invoke Theorem 2.2. Section 8.4 gives the dependency data, and Appendix A gives the Lean names of the lemmas above. The exposition of this section follows the formal proof step by step.

4. ORDINARY VERSUS ACCELERATED PREDECESSORS

One step of T is one step of C_+ on even inputs and two steps of C_+ on odd inputs, since $3n + 1$ is even when n is odd. A T -orbit therefore visits every value of the C_+ -orbit except the outputs $3n + 1$ of odd steps. All of these are $\equiv 4 \pmod{6}$.

Lemma 4.1. *Let $a, n \in \mathbb{Z}$.*

- (i) *If $n \rightsquigarrow_T a$ then $n \rightsquigarrow_{C_+} a$.*
- (ii) *If $a \not\equiv 4 \pmod{6}$ and $n \rightsquigarrow_{C_+} a$, then $n \rightsquigarrow_T a$.*
- (iii) *$T(2a) = a$, so $n \rightsquigarrow_T 2a$ implies $n \rightsquigarrow_T a$. If $a \equiv 4 \pmod{6}$ then $2a \equiv 2 \pmod{6}$, and if $3 \nmid a$ then $3 \nmid 2a$.*

Proof. (i) is the first sentence of this section. For (ii) we use strong induction on the length t of a witness $C_+^t(n) = a$. For $t = 0$ there is nothing to prove. If n is even then $C_+(n) = T(n)$. If n is odd then $C_+(n) = 3n + 1 \equiv 4 \pmod{6} \neq a$, so $t \geq 2$ and $C_+^2(n) = T(n)$. In both cases the induction hypothesis applies to $T(n)$. (iii) is immediate. $\square$

The class $a \equiv 4 \pmod{6}$ is where the two predecessor sets can differ. For such a , the odd integer $m = (a - 1)/3$ satisfies $C_+(m) = a$, while $T(m) = a/2$ steps over a . The sets need not actually differ. For example $-68 \equiv 4 \pmod{6}$ lies on a T -cycle, and there every ordinary predecessor is an accelerated one. The inclusions $P(2a) \subseteq P_T(a) \subseteq P(a)$ for positive a are recorded in [6, §1.2, (1.3)], together with the remark that Theorem 2.2, applied to $2a$, gives positive density for T . For positive targets the next proposition is that observation. We use the finer Lemma 4.1(ii) off the class $4 \pmod{6}$, which is how the formal proof proceeds.

Proposition 4.2 (Positive targets). *Let $a \geq 1$ with $3 \nmid a$. There are $c > 0$ and X_0 such that $\#\{0 < n < X : n \rightsquigarrow_T a\} \geq cX$ for every integer $X \geq X_0$.*

Proof. If $a \not\equiv 4 \pmod{6}$, Lemma 4.1(ii) puts $P(a)$ inside the set counted, and Theorem 2.2 applies. If $a \equiv 4 \pmod{6}$, apply Theorem 2.2 to $2a$. Lemma 4.1(ii) applies to $2a \equiv 2 \pmod{6}$, and Lemma 4.1(iii) transfers from $2a$ to a . $\square$

5. NEGATIVE TARGETS

Lemma 5.1 (Conjugacy). *For every integer $m \geq 1$, $C_+(-m) = -C_-(m)$. Consequently $C_+^k(-m) = -C_-^k(m)$ for every $k \geq 0$, and $m \rightsquigarrow_{C_-} a$ implies $-m \rightsquigarrow_{C_+} -a$.*

Proof. The integers m and $-m$ have the same parity. If m is even, $C_+(-m) = -m/2 = -C_-(m)$. If m is odd, $C_+(-m) = -3m + 1 = -(3m - 1) = -C_-(m)$. Since C_- maps positive integers to positive integers, the identity iterates. $\square$

In other words, the diagram

$$\begin{array}{ccc} \{m \geq 1\} & \xrightarrow{C_-} & \{m \geq 1\} \\ m \mapsto -m \downarrow & & \downarrow m \mapsto -m \\ \{n \leq -1\} & \xrightarrow{C_+} & \{n \leq -1\} \end{array}$$

commutes, and C_+ maps negative integers to negative integers. The $3x + 1$ dynamics on the negative integers is exactly the $3x - 1$ dynamics on the positive integers. Its cycles are the negatives of the $3x - 1$ cycles, and its predecessor sets are the negatives of the $3x - 1$ predecessor sets. Theorem 2.2 says nothing about them. Theorem C is precisely the statement needed.

Proposition 5.2 (Negative targets). *Let $a \geq 1$ with $3 \nmid a$. There are $c > 0$ and X_0 such that $\#\{n \in \mathbb{Z} : -X < n < 0, n \rightsquigarrow_T -a\} \geq cX$ for every integer $X \geq X_0$.*

Proof. By Lemma 5.1, $m \mapsto -m$ maps $\{1 \leq m < X : m \rightsquigarrow_{C_-} a\}$ injectively into the negative C_+ -predecessors of $-a$ in $(-X, 0)$. If $-a \not\equiv 4 \pmod{6}$, these are T -predecessors by Lemma 4.1(ii), and Theorem C gives the bound. If $-a \equiv 4 \pmod{6}$, apply Theorem C to $2a$. Then $-2a \equiv 2 \pmod{6}$, so Lemma 4.1(ii) applies at $-2a$, and Lemma 4.1(iii) transfers from $-2a$ to $-a$. $\square$

The mod-6 correction survives the conjugation. The conjugacy relates the *ordinary* maps, and Theorem C counts ordinary predecessors. The passage to T therefore happens on the $3x + 1$ side, with the same exceptional class.

6. PROOF OF CONJECTURE A

Fix $a \in \mathbb{Z}$ with $3 \nmid a$; in particular $a \neq 0$. For an integer $X \geq 0$ let

$$\Pi_a(X) = \#\{n \in \mathbb{Z} : n \neq 0, |n| \leq X, n \rightsquigarrow_T a\}.$$

Proposition 6.1 (Signed eventual density). *There are $c > 0$ and X_0 with $\Pi_a(X) \geq cX$ for every integer $X \geq X_0$.*

Proof. If $a > 0$, the counted set of Proposition 4.2 lies inside the set counted by $\Pi_a(X)$. If $a < 0$, apply Proposition 5.2 to $|a|$; its counted set also lies inside. $\square$

The printed form of Conjecture A asks for more than an eventual bound. It asks for a bound at every $x \geq |a|$. The gap is closed by the target itself.

Lemma 6.2 (The target counts itself). *If $X \geq |a|$ then $\Pi_a(X) \geq 1$.*

Proof. $a \neq 0$, $|a| \leq X$, and $a \rightsquigarrow_T a$ with $k = 0$. $\square$

Proposition 6.3. *There is $c' > 0$ with $\Pi_a(X) \geq c'X$ for every integer $X \geq |a|$, and $\Pi_a(\lfloor x \rfloor) \geq \frac{c'}{2}x$ for every real $x \geq |a|$.*

Proof. Take c, X_0 from Proposition 6.1 and $c' = \min\{c, 1/(X_0 + 1)\}$. For $X \geq X_0$, $\Pi_a(X) \geq cX \geq c'X$. For $|a| \leq X < X_0$, Lemma 6.2 gives $\Pi_a(X) \geq 1 \geq X/(X_0 + 1) \geq c'X$. For real $x \geq |a| \geq 1$ we have $\lfloor x \rfloor \geq |a|$ and $x \leq 2\lfloor x \rfloor$. $\square$

It remains to compare Π_a with the literal count (1). Since $T(0) = 0$, the integer 0 reaches only 0. As $a \neq 0$, excluding $n = 0$ changes nothing, and so for every real $x \geq 0$

$$(4) \quad \pi_a(x) = \Pi_a(\lfloor x \rfloor).$$

Proof of Theorem A. Combine (4) with Proposition 6.3, and take $c_a = c'/2$. $\square$

Three features of the printed statement are used. The window $|n| \leq x$ is closed, so $x = |a|$ is covered. The case $k = 0$ is allowed in (1), so the target is its own predecessor. And the threshold is $x \geq |a|$, which cannot be lowered to $x \geq 1$: $\pi_7(1) = 0$, because T fixes 0 and -1 and the orbit of 1 is the cycle $\{1, 2\}$. For this reason the paraphrase of Conjecture A in [4, after Conjecture 2.1], which has “ $x \geq 1$ ” in place of “ $x \geq |a|$ ”, cannot hold as stated. The form proved here is the one printed in [1].

7. THE GROWTH EXPONENT CONJECTURE

Proof of Theorem B. The window $[-x, x]$ contains $2\lfloor x \rfloor + 1$ integers, so $\pi_a(x) \leq 2x + 1 \leq 3x$ for $x \geq 1$. With c_a from Theorem A, for $x \geq \max\{|a|, 2\}$ we have $c_ax \leq \pi_a(x) \leq 3x$ and $\log x > 0$, so

$$1 + \frac{\log c_a}{\log x} \leq \frac{\log \pi_a(x)}{\log x} \leq 1 + \frac{\log 3}{\log x}.$$

Both bounds tend to 1. Hence $\log \pi_a(x)/\log x \rightarrow 1$, and in particular $\eta_3^+(a) = \eta_3^-(a) = 1$. $\square$

The count is over $\mathbb{Z}$, so the trivial upper bound is $2x + 1$ rather than $x + 1$. Only its linear growth matters.

8. FORMAL VERIFICATION

This section describes how the proof above was certified in the Lean development [3].

8.1. Environment and headline theorems. The formalization uses Lean v4.30.0-rc2 and Mathlib at commit 5450b53e, the versions pinned by the upstream packages of [5, 6]. It has 464 modules. Of these, 388 are the upstream sources, included without modification: 377 from [5] and 11 from [6]. The remaining 76 are new. They consist of 71 modules for Theorem C, the positive-target and integer assemblies, the growth exponent file and two audit modules. The headline theorems are:

Theorem A	ALConjectureAZ.conjectureA_AL
Theorem B, limit form	ALConjectureAZ.tendsto_growthRatio
Theorem B, as conjectured	ALConjectureAZ.growthExponentConjecture
Theorem C	ThreeXMinusOne.predecessors_positive_lower_density
Theorem 2.2 (input)	CollatzPredecessorDensity.predecessors_positive_lower_density

Each of them, and Proposition 4.2 (ALConjectureA.conjectureA), depends only on the axioms `propext`, `Classical.choice` and `Quot.sound`. No file contains `sorry`, a new axiom or `native_decide`.

8.2. Statement fidelity. Reading a formal statement and comparing it with a printed one by eye is the weakest link in any claim of this kind, so we do not rely on it. An audit file restates both conjectures using only core Lean and Mathlib notions, namely function iteration, integer arithmetic, set cardinality, the real logarithm and filter limits. The restatements refer to no definition of the project. Each is then discharged by the corresponding theorem as a bare term. If any definition in the chain differed from what is written, or if a theorem had acquired an extra hypothesis, these declarations would fail to elaborate.

```

/-- Applegate-Lagarias Conjecture A, written with no project definitions. -/
example :
  ∀ a : ℤ, ¬ (3 ∣ a) →
    ∃ c : ℝ, 0 < c ∧ ∀ x : ℝ, |(a : ℝ)| ≤ x →
      c * x ≤ ({ n : ℤ ∣ |(n : ℝ)| ≤ x ∧ ∃ k : ℕ,
        (fun m : ℤ => if m % 2 = 0 then m / 2
          else (3 * m + 1) / 2)^[k] n = a }.ncard : ℝ) :=
  ALConjectureAZ.conjectureA_AL

/-- The growth exponent conjecture, likewise. -/
example :
  ∀ a : ℤ, ¬ (3 ∣ a) →
    Filter.liminf (fun x : ℝ =>
      Real.log ({ n : ℤ ∣ |(n : ℝ)| ≤ x ∧ ∃ k : ℕ,
        (fun m : ℤ => if m % 2 = 0 then m / 2
          else (3 * m + 1) / 2)^[k] n = a }.ncard : ℝ)
        / Real.log x) Filter.atTop = 1 ∧
    Filter.limsup (fun x : ℝ =>
      Real.log ({ n : ℤ ∣ |(n : ℝ)| ≤ x ∧ ∃ k : ℕ,
        (fun m : ℤ => if m % 2 = 0 then m / 2
          else (3 * m + 1) / 2)^[k] n = a }.ncard : ℝ)
        / Real.log x) Filter.atTop = 1 :=
  ALConjectureAZ.growthExponentConjecture

```

Three Lean conventions could make a statement of this shape mean less than it appears to. We checked each one.

- *Integer division.* The map divides m by 2 only when m is even, and $3m + 1$ by 2 only when m is odd, when $3m + 1$ is even. Both divisions are exact, so the rounding convention of integer division is irrelevant. The function written is exactly T .
- *Cardinality of infinite sets.* `Set.ncard` is 0 on infinite sets, which could make a lower bound hold vacuously. Here the set lies in $[-x, x]$, and the proof identifies it with an explicit finite count, which is (4).
- *Junk values.* In Mathlib $\log 0 = 0$, and $\liminf$ and $\limsup$ of a real function that is not bounded take a default value. Neither convention is used: $\pi_a(x) \geq 1$ for $x \geq |a|$, and the proof establishes genuine convergence, from which both one-sided limits follow.

8.3. Closedness checks. A second command in the audit file walks the transitive dependencies of eight roots. These are the four successive forms of Theorem A, the two forms of Theorem B, Theorem C and Proposition 4.2. The command fails the build if it finds an axiom outside the three standard ones, or a free variable in the type of a root. A further check fails the build if any declaration asserting the universal $3x + 1$ conjecture is present. The upstream package deleted the unproved universal statement from a file it includes, and the check keeps it absent.

8.4. Dependency structure. A transitive-closure computation over the Lean environment confirms the architecture of Figure 1. Theorem A depends on both Theorem 2.2 and Theorem C, and Proposition 4.2 depends on Theorem 2.2 but not on Theorem C. Theorem C does not depend on Theorem 2.2, nor on the upstream seed theorem, $3x + 1$ counting theorem or telescoped charge inequality. Its dependency closure has 43,403 declarations: 549 from the new modules, 3,970 from the upstream packages, and the rest from Mathlib and the Lean core. The new modules use 119 upstream theorems directly. They include the mixing estimate, the residue-selection lemma, the word combinatorics, the scale-coverage arithmetic and the non-return lemma for arbitrary maps. A few $3x + 1$ residue identities also enter, through the substitution of Lemma 3.2.

8.5. Independent kernel replay. `#print axioms` and the dependency walk inspect the environment produced by the elaborator. They do not re-check the stored proof terms. The toolchain’s `leanchecker` with the option `--fresh` does: it re-reads the compiled files and replays every declaration, including those imported from Mathlib and from the upstream packages, into an empty kernel environment. We replayed the module `GrowthExponent`, whose import closure contains Theorems A, B and C and Theorem 2.2, from a fresh clone of the repository. `leanchecker --fresh` exited with status 0 after 12 minutes on an 8 GB laptop. Earlier replays of `ConjectureAZ` and `GrowthExponent` in the development tree, on the same sources, also exited with status 0. The upstream releases record no replay of this kind [5, 6]; the replay reported here re-checks their declarations as well. The replay is informative. It rejects a missing module (exit status 1) and a truncated object file (exit status 139).

8.6. Provenance and reproducibility. A script recomputes the SHA-256 hash of each of the 388 included upstream files. Each matches the hash in the published upstream manifests. The 11 files from [6] are also byte-identical to those of its current source release (version 1.1). The repository builds from a fresh clone with `lake exe cache get` followed by `lake build`. On an 8 GB laptop the cold build of all 464 modules took about 4 hours 15 minutes. It completed without errors and printed the axiom reports and both audit messages.

8.7. Trusted base, and what the formalization does not cover. Accepting Theorems A and B on the strength of the formal proof means trusting the Lean kernel, the three standard axioms, and the Mathlib definitions that occur in the restated statements. It does not require trusting any definition, lemma or proof of this project or of the upstream packages. The formalization certifies the theorems, not this exposition. Section 3 describes the formal proof of Theorem C and was written from it, but the correspondence is not mechanically checked. The same holds for the manuscripts [5, 6] and the formal proof they describe. During development, three lemmas of the $3x - 1$ proof compiled and yet had the wrong statement for the lemma that used them. All three were found by reading the consuming statement. None survives in the final proof, which is checked end to end. A successful build certifies that each term proves its stated type. It does not certify that an intermediate statement is the one a human reader would expect.

9. CONSTANTS AND LIMITATIONS

9.1. The constants are ineffective. Theorem A is existential in c_a . The proof determines a witness, and the witness is extremely small. For positive targets the constant is $c = 3/(256 R 3^m)$ with $m = 132RC + 1$ [6, (4.3), (4.6)]. Here $C > 10^{42383}$ is the explicit mixing coefficient of [5, Lemma 8.1], and $R \geq 16^{2^{80}}$ is the seed. For negative targets the constant of Section 3 has the same shape. The threshold step of Proposition 6.3 then replaces the constant by $1/(X_0 + 1)$, where X_0 is a tower-type function of R . That step is the largest single loss.

The witness is therefore bounded above by an absolute constant, of order $3^{-132 \cdot 16^{2^{80}} C}$, because it decreases in R and $R \geq 16^{2^{80}}$. No positive lower bound on it can be extracted, because the seed comes from an existential with no upper bound in terms of a [6, Remark 4.3]. Some dependence on a is in any case unavoidable. Any admissible constant satisfies $c_a \leq \pi_a(|a|)/|a|$, and a direct computation gives $\pi_{100}(100) = \pi_{10^4}(10^4) = 1$.

9.2. What is and is not established.

- Theorems A, B and C are formally proved and independently replayed by the Lean kernel, together with Theorem 2.2.
- The human-readable proof of Theorem 2.2 in [5, 6] and our account of Theorem C in Section 3 have not been independently refereed. The provenance of [5, 6] is described in Section 2.3. A kernel check establishes that the formal derivations are valid. Human review of the mathematics remains a separate and worthwhile step.
- The proof gives no bound of the form $c_a \geq c/|a|$ with c independent of a . The obstacle is structural: the argument works at a seed far above the target, not at the target.
- Nothing is claimed about the $3x + 1$ conjecture itself. Theorem A for $a = 1$ says only that a positive proportion of the integers in $[-x, x]$ reach 1.

10. FURTHER DIRECTIONS

Applegate and Lagarias state two further conjectures about the tree of preimages. Conjecture B asks that the number $n_k(a)$ of n with $T^{(k)}(n) = a$ satisfy $n_k(a) = (4/3)^{k(1+o(1))}$. Conjecture C and the stronger Conjecture C' concern the extreme tree sizes over all targets [1, §§1–2]. Neither follows from the methods here, which count predecessors at all depths at once. Within the present method, an effective version of Theorem A, with c_a bounded below by an explicit function of a , would require an explicit upper bound on the seed in terms of a . We intend to return to the distribution of $3x + 1$ trees in separate work.

CODE AND DATA AVAILABILITY

The complete Lean development is licensed under Apache 2.0 [3]. It includes the upstream sources with their licences and manifests, the audit file and a provenance script. Its address is not given in this version of the manuscript. Appendix A maps the results of this paper to Lean declarations.

ACKNOWLEDGEMENTS

The author thanks Lech Mazur for releasing the formal sources of [5, 6] under an open licence, which made this work possible, and the Mathlib community for the library on which it rests.

USE OF GENERATIVE AI

Generative AI (Anthropic’s Claude, used through Claude Code) was used extensively to develop the Lean formalization of Theorem C and of the reductions in Sections 4–7, and to draft this manuscript. The upstream packages [5, 6] were likewise developed with substantial AI assistance, as their manuscripts state. The formal proof is checked as described in Section 8. The author takes full responsibility for the content of this paper.

APPENDIX A. RESULTS AND LEAN DECLARATIONS

Result	Lean declaration
Theorem 2.2 (input)	<code>CollatzPredecessorDensity.predecessors_positive_lower_density</code>
Lemma 3.1	<code>ThreeXMinusOne.taoAffListM_oddNat_decode</code>
Lemma 3.2, exactness	<code>ThreeXMinusOne.taoAffineOffsetZModM_eq_iff_cleared</code>
Corollary 3.3(b)	<code>ThreeXMinusOne.sum_unitIncidenceM_mark_eq_filteredKernel,</code> <code>ThreeXMinusOne.ndTernaryUniformMean_neg</code>
(R1)	<code>ThreeXMinusOne.prefixMassM_succ_eq_mul</code>
(R3)	<code>ThreeXMinusOne.crossBoundsM_of_boundedOvershoot</code>
Lemma 3.4	<code>ThreeXMinusOne.threePow_depthM_le,</code> <code>ThreeXMinusOne.sourceM_mul_atomM_le_root_sharp</code>
Lemma 3.5	<code>ThreeXMinusOne.generation_product_le</code>
Corollary 3.6(i)	<code>ThreeXMinusOne.fullTerminalM_source_mul_weight_le_owner,</code> <code>ThreeXMinusOne.generationExcessProductM_le</code>
Lemma 3.7	<code>ThreeXMinusOne.exists_large_nonreturning_predecessorM_in_residue</code>
Seed class $-y$	<code>ThreeXMinusOne.exists_markM_seed</code>
Surviving mass $\geq 1/4$	<code>ThreeXMinusOne.exists_frozenSeedStateM</code>
Counting, window $[X, 64X)$	<code>ThreeXMinusOne.generalTargetM_publicCount_from_mass</code>
Theorem C	<code>ThreeXMinusOne.predecessors_positive_lower_density</code>
Lemma 4.1	<code>ALConjectureAZ.reachesSyrZ_iff,</code> <code>ALConjectureAZ.reachesSyrZ_two_mul,</code> <code>ALConjectureAZ.two_mul_mod_six_Z</code>
Proposition 4.2	<code>ALConjectureA.conjectureA</code>
Lemma 5.1	<code>ALConjectureAZ.neg_colZ, ALConjectureAZ.colZ_iterate_neg</code>
Proposition 5.2	<code>ALConjectureAZ.conjectureAZ_neg_of_predecessorDensity</code>
Proposition 6.1	<code>ALConjectureAZ.conjectureAZ</code>
Lemma 6.2	<code>ALConjectureAZ.one_le_piAZ</code>
Proposition 6.3	<code>ALConjectureAZ.conjectureAZ', ALConjectureAZ.conjectureAZReal</code>
Equation (4)	<code>ALConjectureAZ.piAL_eq_piAZ</code>
Theorem A	<code>ALConjectureAZ.conjectureA_AL</code>
Theorem B	<code>ALConjectureAZ.tendsto_growthRatio,</code> <code>ALConjectureAZ.growthExponentConjecture</code>
Audit	<code>Verification/ConjectureAExact.lean</code>

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